IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

Lauren Bleyaert
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IBM Data Science Professional Certificate

# Outline

# Executive Summary

- Worked with SpaceX launch data to better understand how launch success might be determined
- Launch success was found to be related to a variety of factors
  - Orbit type
  - Payload mass
  - Date (improvement over time)
- Launch site locations appear to be dependent on a variety of factors
  - Proximity to landmarks such as coastline, railways, cities

# Introduction

- The goal of the project is to determine when the first stage of Falcon 9 will land successfully
  - Reusing a launch's first stage reduces the cost of the launch significantly
  - If we can determine when the first stage gets reused, we can more easily predict launch pricing

Section 1

# Methodology

# Methodology

## Executive Summary

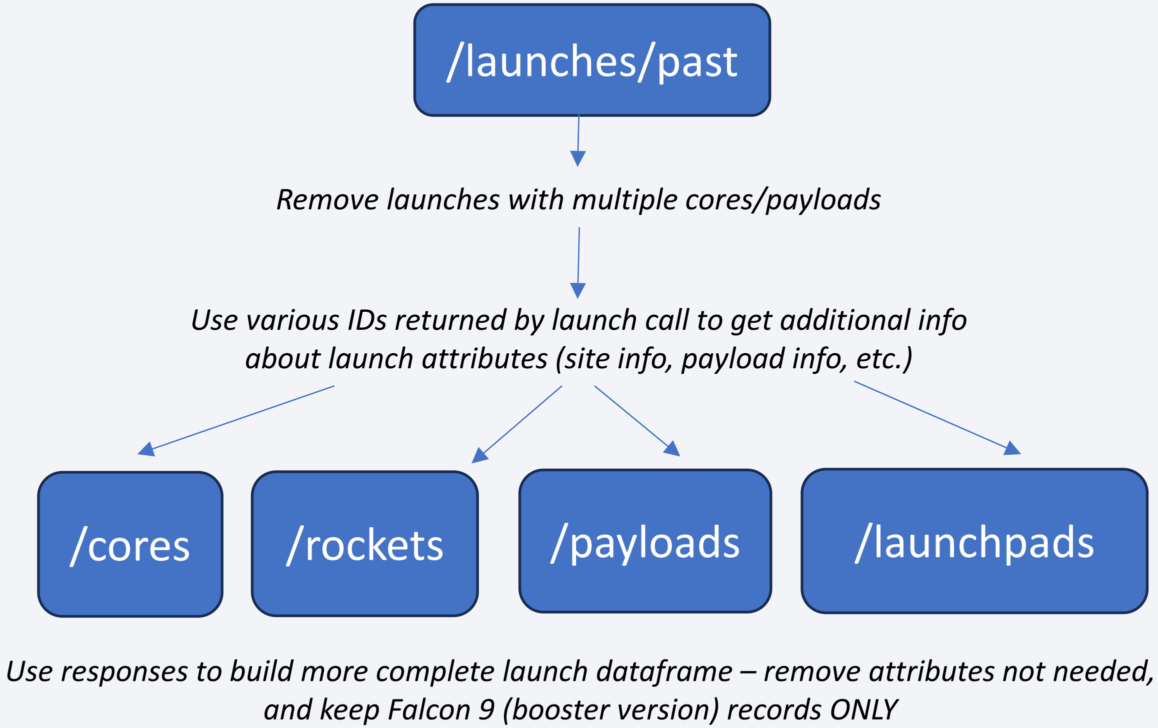
- Data collection methodology:
  - SpaceX API and Wikipedia Webscraping
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

# Data Collection

- SpaceX API
  - Performed API call for past **launch data** from https://api.spacexdata.com/v4/launches/past
  - Made additional calls for each returned launch to gather additional launch attributes such as **booster name**, **payload mass**, **launch site, orbit, landing type,** and **outcome**
- Wikipedia
  - Collected Falcon 9 historical launch data from Wikipedia page using web scraping tools: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

# Data Collection – SpaceX API

- Performed API call (blue square in flow chart) for past launch data
- Made additional calls (blue squares) for each returned launch to gather additional launch attributes such as **booster name**, **payload mass**, **launch site**, and **outcome**
  - API endpoint: https://api.spacexdata.com/v4/
- GitHub URL – Data Collection




| Index | Flight Number | Date | Booster Version | PayloadMass | Outcome | ..... |
|---|---|---|---|---|---|---|
| | | | | | | |

# Data Collection - Scraping

- Performed HTTP GET request to pull HTML from this Wikipedia webpage
- Used **BeautifulSoup** Python package to process HTML and prepared Pandas Dataframe with relevant dataset
- GitHub URL - Webscraping

HTTP: GET()

HTML response

*Process HTML response with **BeautifulSoup** package:*

1. *Find relevant **table html object***
2. *Extract column names from table headers – create empty dictionary object with the needed cols as keys*
3. *Parse table body and fill dictionary object with data from each record*
4. *Create Pandas Dataframe from dictionary*

| Index | Flight No | Launch site | PayloadMass | Orbit | Customer | ..... |
|---|---|---|---|---|---|---|
| | | | | | | |

# Data Wrangling

- Performed Data Wrangling and Exploratory Data Analysis on retrieved data
- Observed the spread of launches across site, orbit, and outcome
- Categorized ‘good’ vs ‘bad’ outcomes into a binary column
- GitHub URL – Data Wrangling

Identified data types of the columns and missing values in columns

Determined number of launches: per site, per orbit, per outcome

```
LaunchSite
CCAFS SLC 40    55
KSC LC 39A      22
VAFB SLC 4E     13
Name: count, dtype: int64
```

```
Orbit
GTO      27
ISS      21
VLEO     14
PO        9
LEO       7
SSO       5
MEO       3
ES-L1     1
HEO       1
SO        1
GEO       1
```

```
Outcome
True ASDS      41
None None      19
True RTLS      14
False ASDS      6
True Ocean      5
False Ocean     2
None ASDS       2
False RTLS      1
```

Identified unsuccessful outcomes

```
{'False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None'}
```

Marked launch records with **bad** outcome as *Class* = 0 (*Class*= 1 for **good** outcomes)
Took the mean of *Class* column to determine success rate of the launches

```
In [39]: df["Class"].mean()

Out[39]: 0.6666666666666666
```

# EDA with Data Visualization

- Used PyPlot from Matplotlib to plot the outcomes (*Class*) using number of attributes in order to identify patterns and relationships:
    - Flight Number and Payload Mass
    - Flight Number and Launch Site
    - Payload and Launch Site
    - Success Rate (from *Class*) and Orbit
    - Flight Number and Orbit Type
    - Orbit Type vs Payload Mass
    - Launch Success Yearly Trend
- GitHub URL – EDA with Data Viz

See **Results** section for charts and descriptions of identified trends

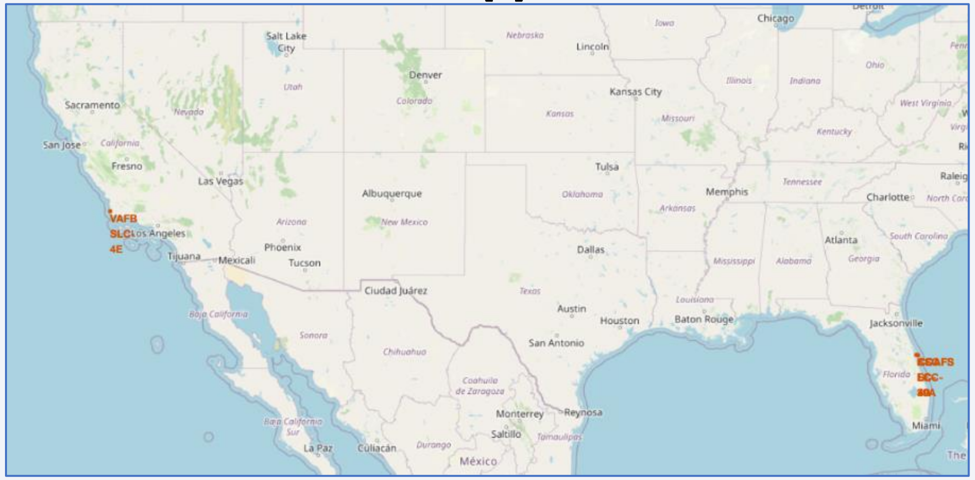
# EDA with SQL

- Created SQL table using SQLite
- SQL Queries performed for further analysis:
  - Identified names of the distinct Launch Sites
  - Summed the Payload Mass for all launches for customer NASA (CRS)
  - Calculated average Payload Mass carried by Booster version F9 v1.1
  - Determined first successful landing date using ground pad
  - Identified boosters that have success in drone ship landings where the Payload Mass was between 4k - 6k (kg)
  - Determined the number of successful/failed missions by outcome
  - Identified boosters that have carried the maximum Payload Mass
  - Pulled the records that had failed drone ship landings in 2015
  - Counted number landings per Landing Outcome between the period of 2010-06-04 and 2017-03-20
- GitHub URL – EDA SQL

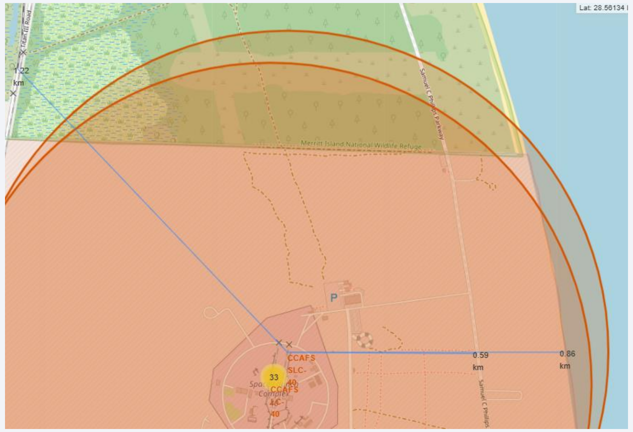
# Interactive Map with Folium

- Created map with markers [1] at locations of the following Space X launch sites:
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- Added marker clusters at each site to show successful (green) and failed (red) launches [2]
- Drew lines and calculated distance to various landmarks [3]:
  - Railway
  - Highway
  - Coastline
- [GitHub URL - Folium]()

[1]




[3]



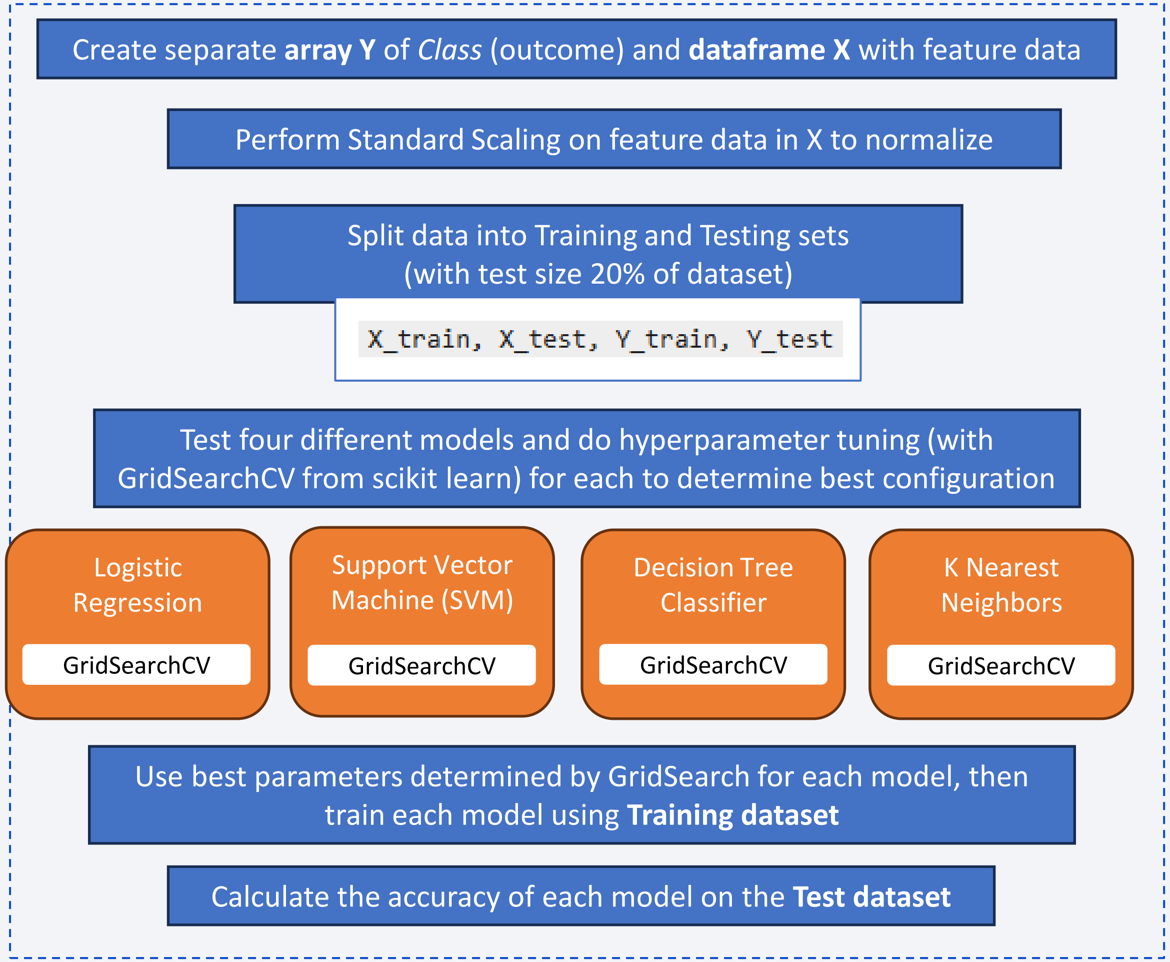
[2]

# Dashboard with Plotly Dash

- Used Plotly library to create a dashboard that shows:
  - % of Successful Launches by Launch Site
  - Success/Failure % by individual Launch Site
  - Correlation between Payload and Success
- Used these visualization to understand how launch success correlated with features such as Launch Site and Payload Mass
- GitHub URL – Plotly Dashboard

# Predictive Analysis (Classification)

- Built models using four different models, determining the best parameters for each model
- Used Train/Test data split to train then evaluate the model
- Evaluated the accuracy of each model option
- [GitHub URL – Predictive Analysis]



# Results

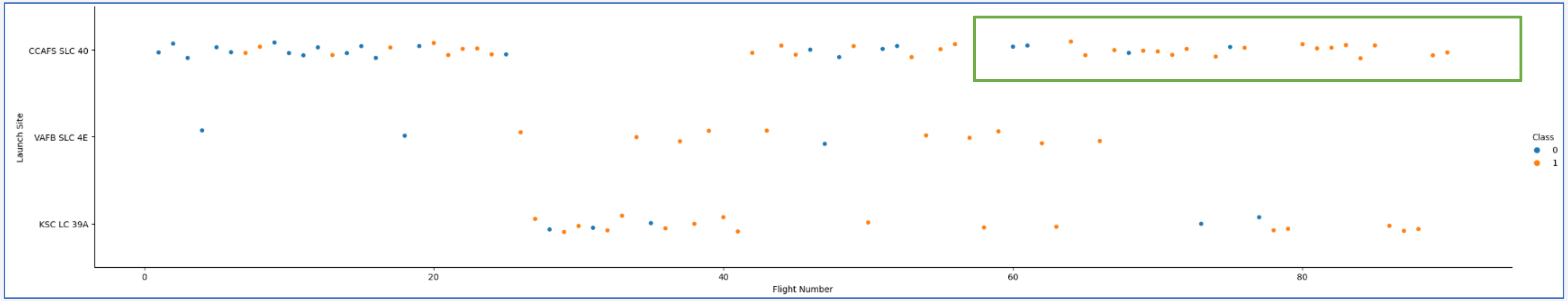
- Launch success generally improved over time
- Launches more frequently done with smaller payloads
- Large payload launches are only done from specific Launch Sites
- Success rates were better for certain orbit types
- Payload seems dependent on the orbit type of the launch
- Various models tested performed with similar accuracy, therefore any could be utilized for prediction
- Launch Site location depends on variety of factors such as proximity to coastline, railways, and cities

Section 2

# Insights drawn from EDA

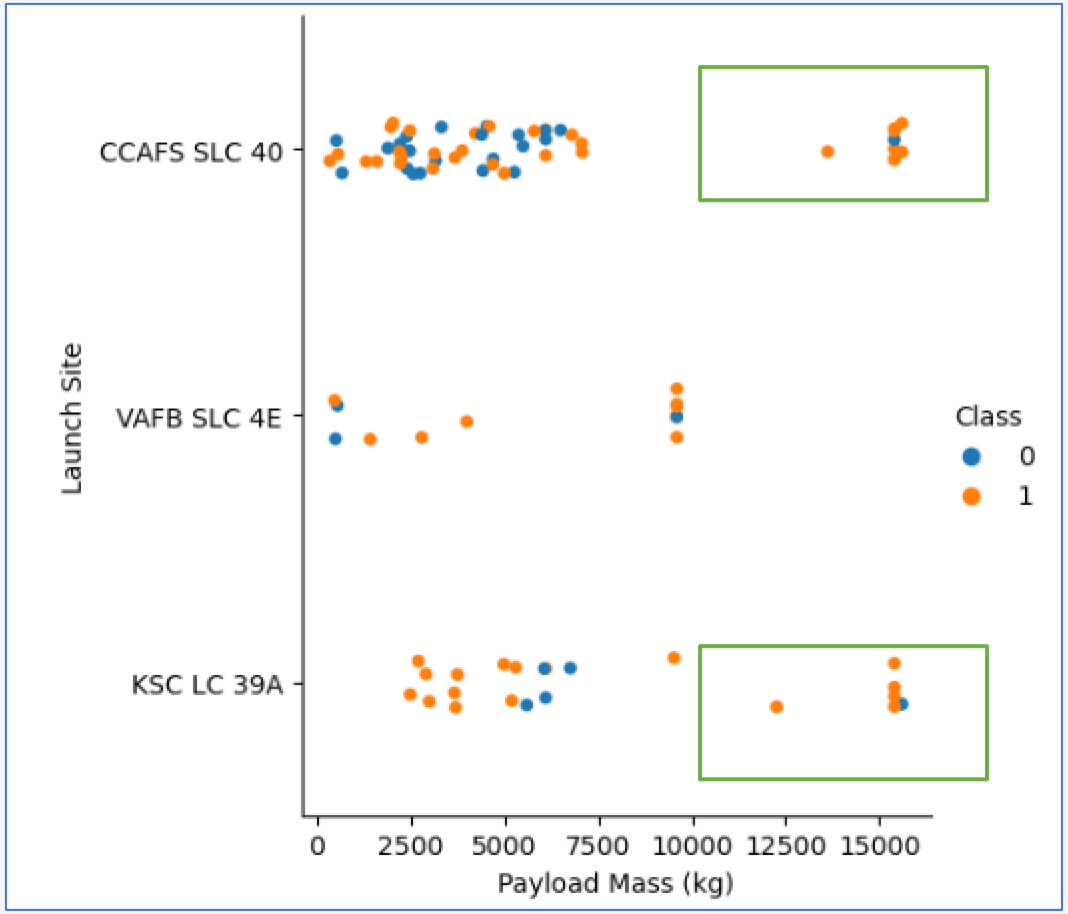
# Flight Number vs. Launch Site

- Observed an **increasing success rate for launches at CCAFS SLC 40** as the flight number increases. So, as time went on and they performed more launches, the success improved.



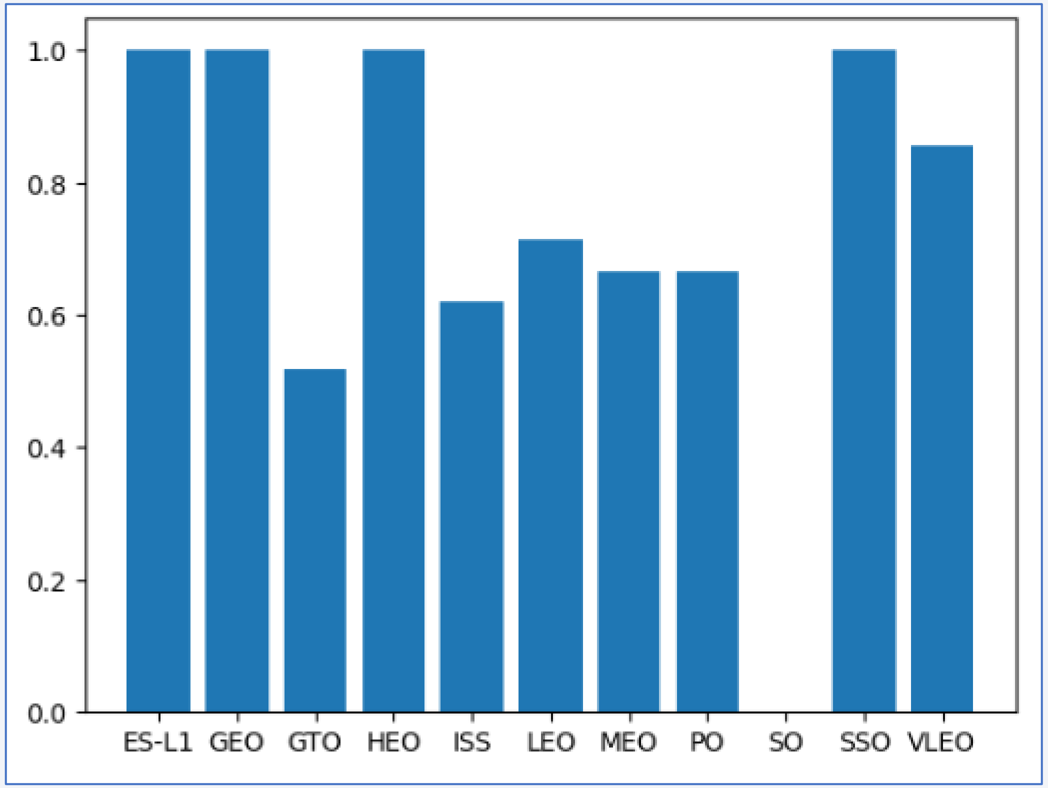
# Payload vs. Launch Site

- Observed that **large payloads were only launched at CCAFS SLC 40 and KSC LC 39A sites**
    - Large payload being greater than 10k kg
    - Site VAFB SLC 4E was only used for smaller payloads



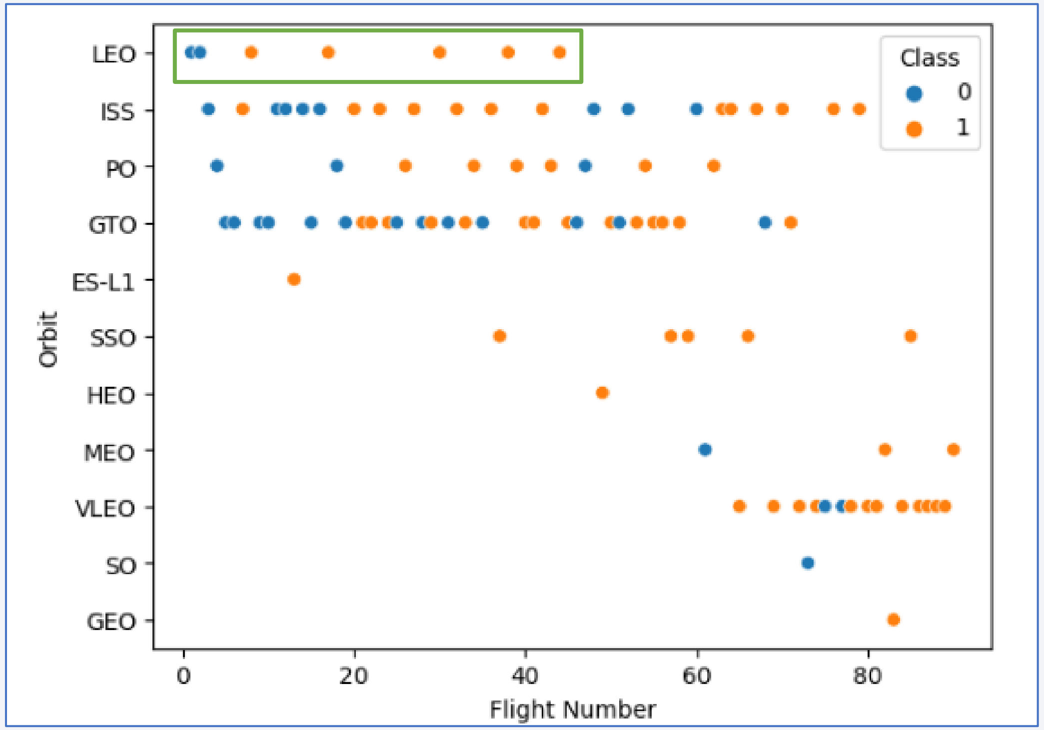
# Success Rate vs. Orbit Type

- Observed that success rates are highest for the following orbits:
    - ES-L1
    - GEO
    - HEO
    - SSO



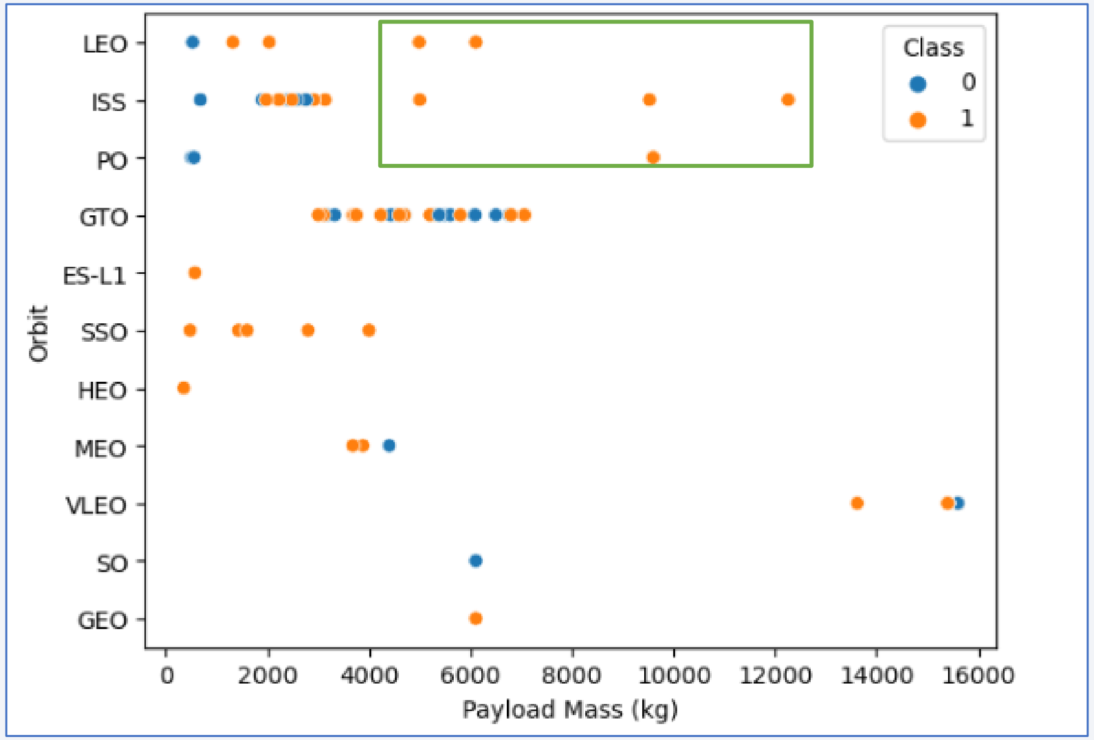
# Flight Number vs. Orbit Type

- Observed that success for LEO appears related to Flight Number
  - improved success as they did more launches
- Other Orbit Types did not appear to have a clear relationship with Flight Number



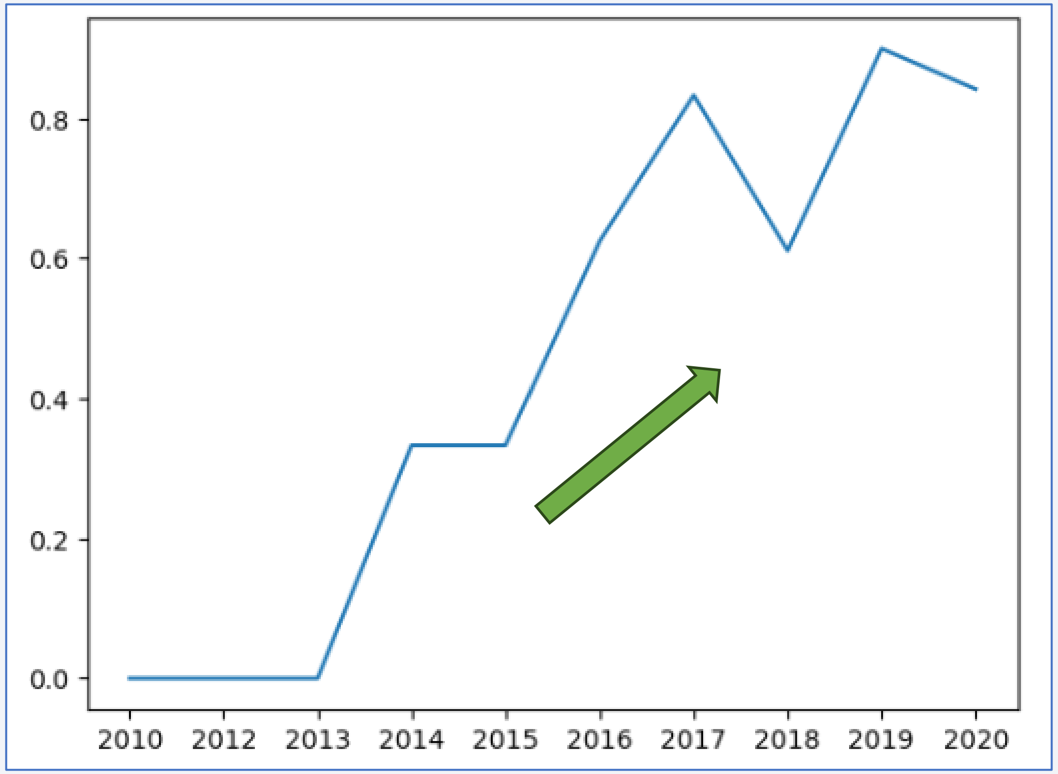
# Payload vs. Orbit Type

- Observed higher successful landing rates for heavy payloads in **PO, LEO, ISS** orbits
- Observed that some Orbit Types never see a large payload (>8k):
  - GTO
  - ES-L1
  - SSO
  - HEO
  - MEO
  - SO
  - GEO



# Launch Success Yearly Trend

- Success rate **generally increasing over time**, with some peaks and valleys between
  - Success rate steady from 2014 to 2015
  - Decreased success rate from 2017 it to 2018



# All Launch Site Names

- Find the names of the unique launch sites
  - There are 4 unique launch sites

```
%sql select DISTINCT Launch_Site FROM SPACEXTABLE
```

```
 * sqlite:///my_data1.db
Done.
```

| Launch_Site |
|---|
| CCAFS LC-40 |
| VAFB SLC-4E |
| KSC LC-39A |
| CCAFS SLC-40 |

# Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
  - Record returned will either be from launch site **CCAFS LC-40** and **CCAFS SLC-40**

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

| Date | Time (UTC) | Booster_Version | Launch_Site | Payload | PAYLOAD_MASS__KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | 18:45:00 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualification Unit | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | 15:43:00 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | 7:44:00 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | 0:35:00 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | 15:10:00 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

# Total Payload Mass

- Calculate the total payload carried by boosters from NASA

```
%sql SELECT Customer, SUM(PAYLOAD_MASS__KG_) as Payload_Total FROM SPACEXTABLE WHERE Customer='NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

| Customer | Payload_Total |
| --- | --- |
| NASA (CRS) | 45596 |

# Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%sql SELECT Booster_Version, AVG(PAYLOAD_MASS__KG_) as Avg_Payload FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%'
```

```
 * sqlite:///my_data1.db
Done.
```

| Booster_Version | Avg_Payload |
|---|---|
| F9 v1.1 B1003 | 2534.6666666666665 |

# First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%sql SELECT MIN(Date) as EarliestSuccessful, Landing_Outcome FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

| EarliestSuccessful | Landing_Outcome |
|---|---|
| 2015-12-22 | Success (ground pad) |

# Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_>4000 and PAYLOAD_MASS__KG_<6000 and Landing_Outcome='Success (drone ship)'
```

```
 * sqlite:///my_data1.db
Done.
```

| Booster_Version |
|---|
| F9 FT B1022 |
| F9 FT B1026 |
| F9 FT B1021.2 |
| F9 FT B1031.2 |

# Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) as MissionCount FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
 * sqlite:///my_data1.db
Done.
```

| Mission_Outcome | MissionCount |
|---|---|
| Failure (in flight) | 1 |
| Success | 98 |
| Success | 1 |
| Success (payload status unclear) | 1 |

# Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

```
 * sqlite:///my_data1.db
Done.
```

| Booster_Version |
|---|
| F9 B5 B1048.4 |
| F9 B5 B1049.4 |
| F9 B5 B1051.3 |
| F9 B5 B1056.4 |
| F9 B5 B1048.5 |
| F9 B5 B1051.4 |
| F9 B5 B1049.5 |
| F9 B5 B1060.2 |
| F9 B5 B1058.3 |
| F9 B5 B1051.6 |
| F9 B5 B1060.3 |
| F9 B5 B1049.7 |

# 2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
SELECT substr(Date,6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND Landing_Outcome='Failu
```

```
 * sqlite:///my_data1.db
Done.
```

| Month | Landing_Outcome | Booster_Version | Launch_Site |
|---|---|---|---|
| 01 | Failure (drone ship) | F9 v1.1 B1012 | CCAFS LC-40 |
| 04 | Failure (drone ship) | F9 v1.1 B1015 | CCAFS LC-40 |

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql SELECT Landing_Outcome, COUNT(*) as Ct FROM SPACEXTABLE WHERE DATE>'2010-06-04' AND DATE<'2017-03-20' GROUP BY Landing_Outcome ORDER BY Ct DESC
```

```
 * sqlite:///my_data1.db
Done.
```

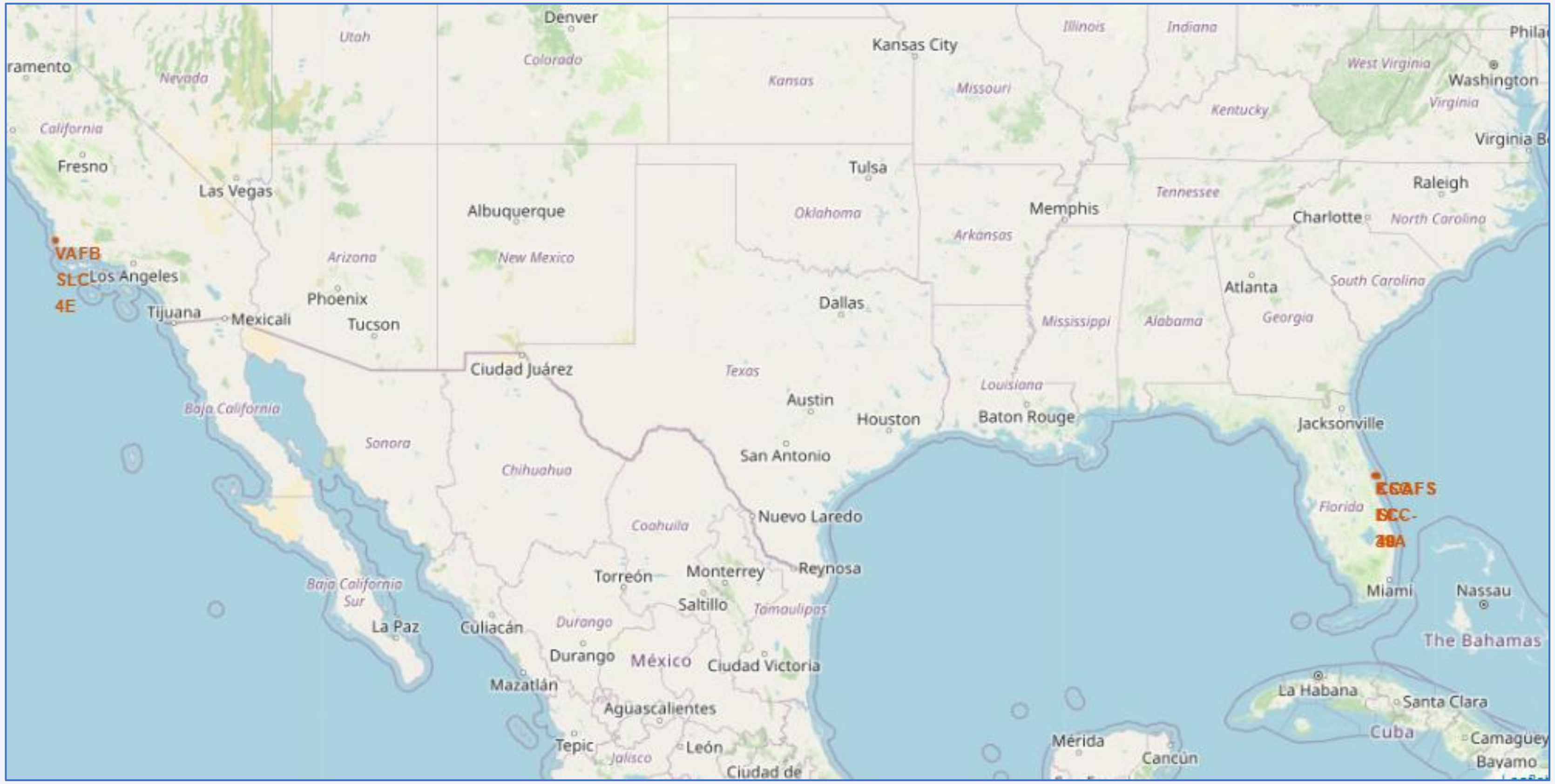
| Landing_Outcome | Ct |
|---|---|
| No attempt | 10 |
| Success (drone ship) | 5 |
| Failure (drone ship) | 5 |
| Success (ground pad) | 3 |
| Controlled (ocean) | 3 |
| Uncontrolled (ocean) | 2 |
| Precluded (drone ship) | 1 |
| Failure (parachute) | 1 |

Section 3

# Launch Sites Proximities Analysis

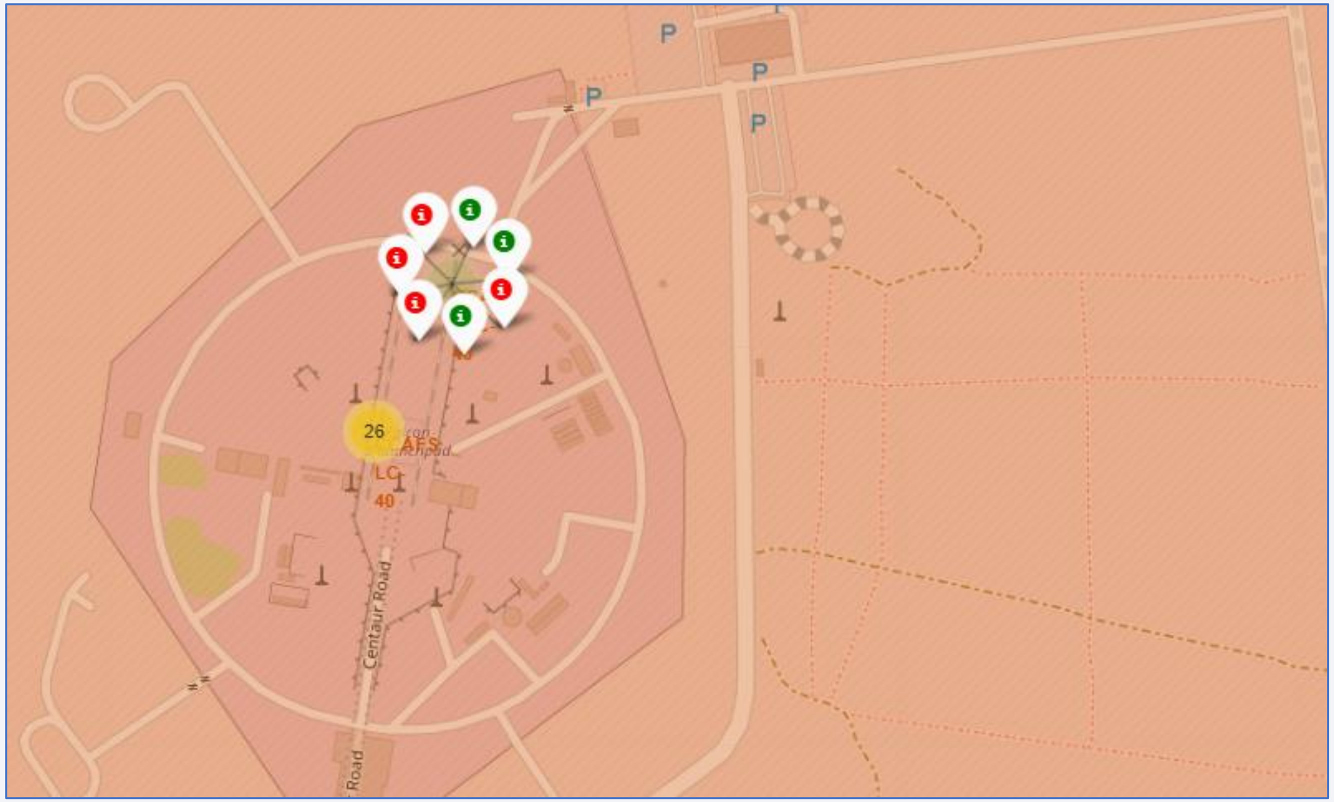
# Map with Launch Site Markers

- Locations marked in orange for following Launch Sites
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- First 3 sites are clustered in Florida



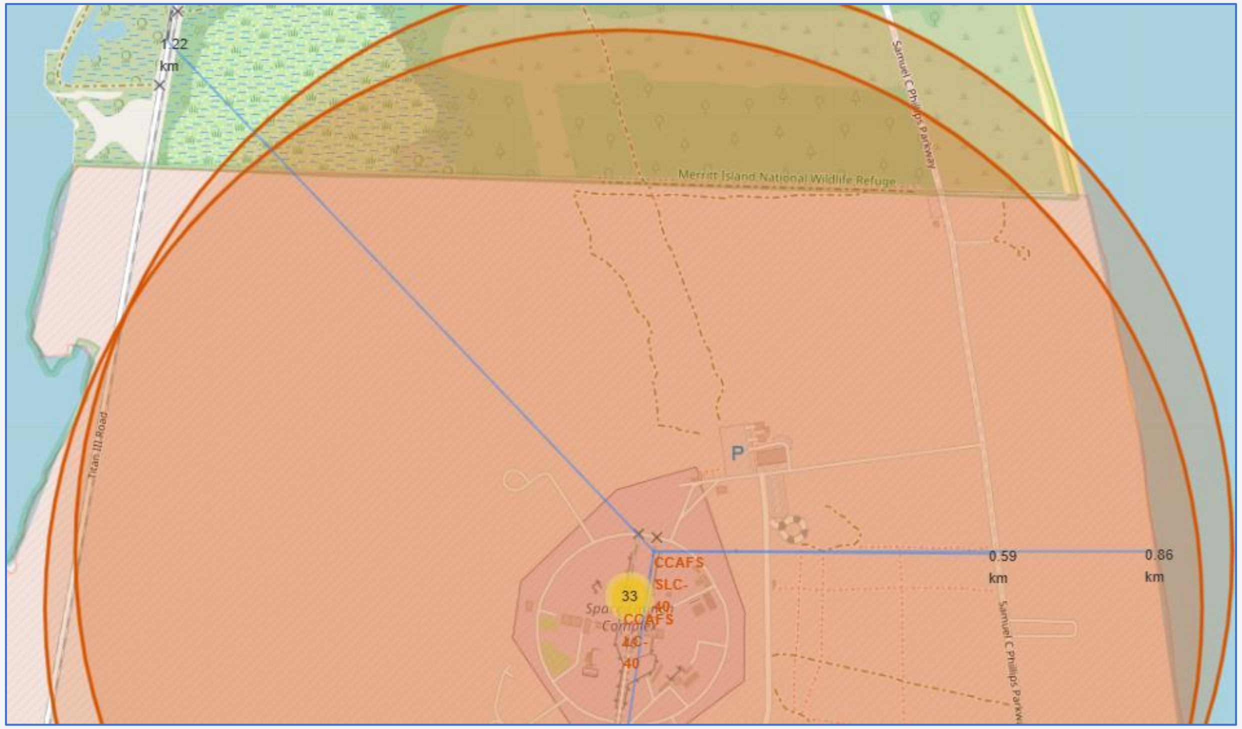
# Launch Results by Outcome

- Folium map zooms in on launch sites and shows indicators for successful (green) and (failed) launches at that site
  - Screenshot shows result markers 7 launches at the CCAFS SLC 40 site



# Distance from Landmarks

- Plotted launch site proximity to various landmarks such as railway, highway, coastline. Distances are calculated and displayed.
    - All launch sites appear to be within a certain proximity this is possibly due to water landings being an option
    - Distance between launch sites and railways appears to be needed, but they are in the general vicinity
    - Appears that a significant distance needed between launch site and cities – this is possibly due to safety concerns

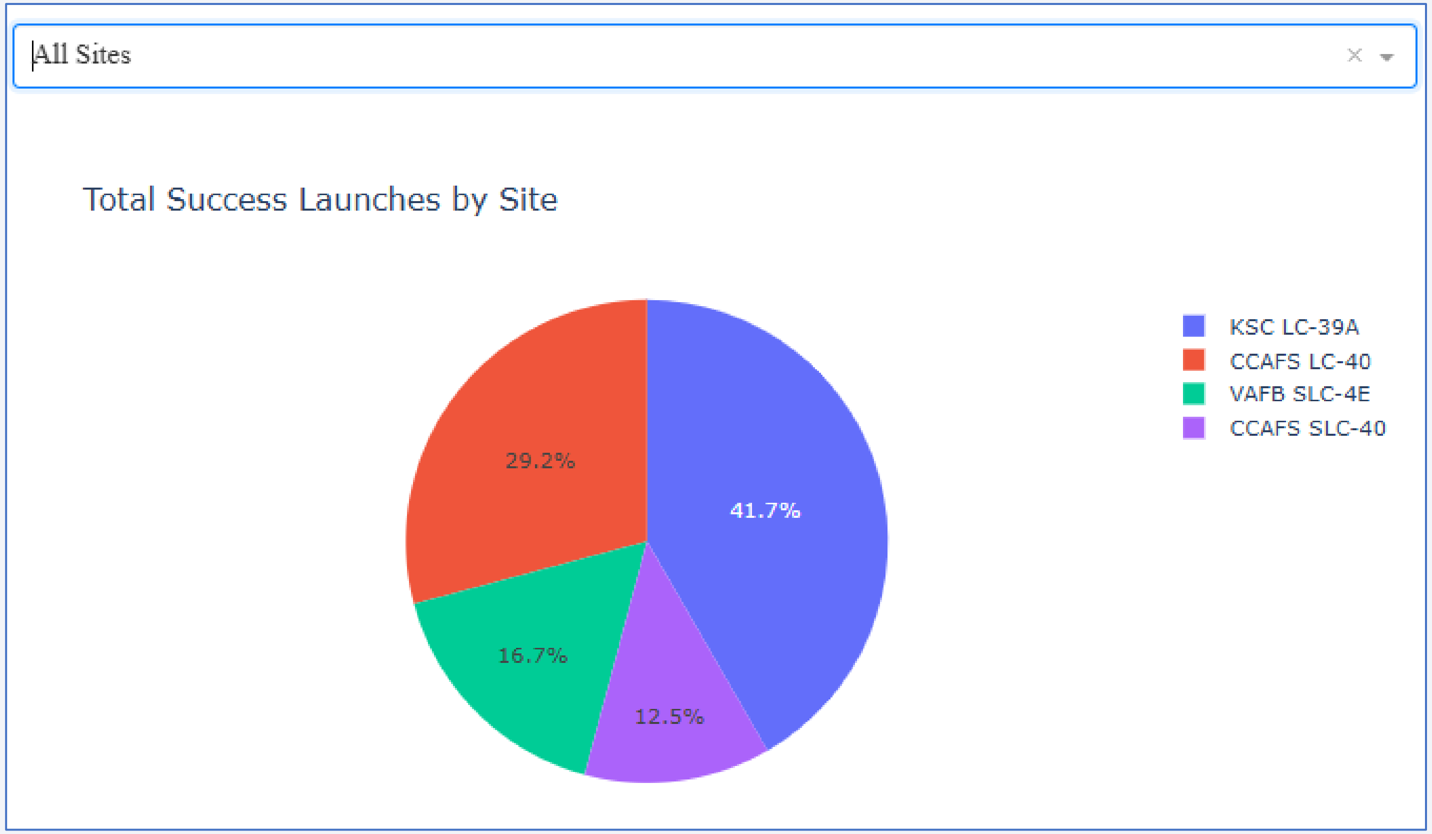


Section 4

# Build a Dashboard with Plotly Dash

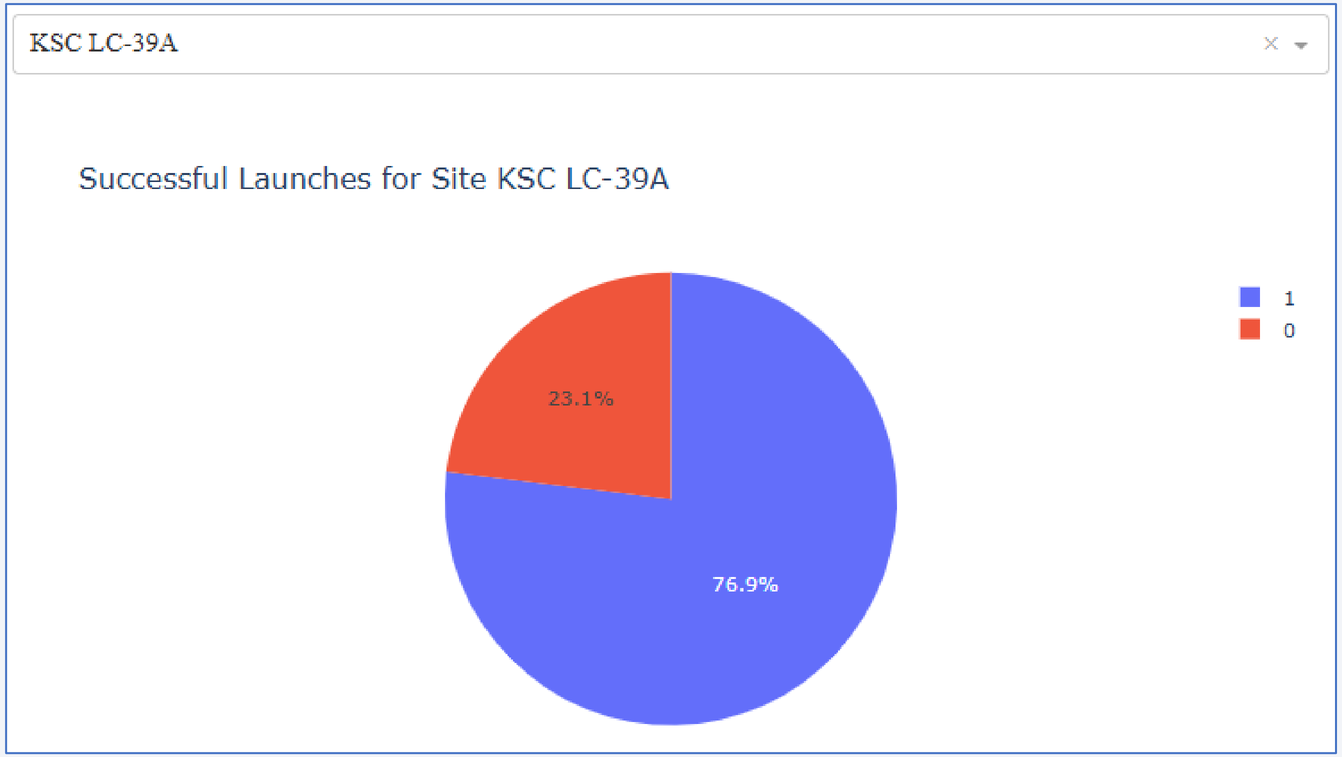
# Successful Launch Results for All Sites

- Visual shows total successful launches broken out by Launch Site
- Site **KSC LC-39A** has most significant amount of successful launches



# Launch Successes and Failures for KSC LC-39A

- Selecting just the site with the most successes **(KSC LC-39A)** allows us to see successes (blue) vs failures (red) for that site
- App allows you to other individual sites to see their successes vs failures



# Payload vs Class (success or failure)

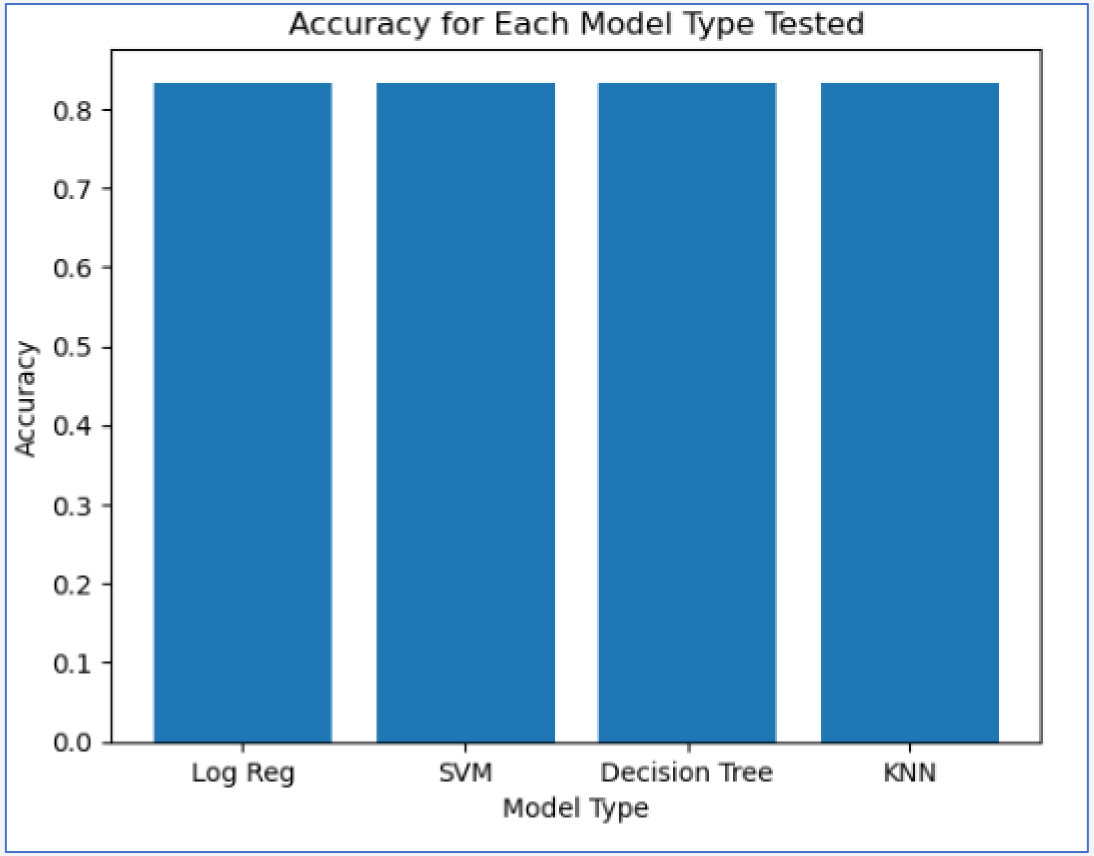
- Select Payload Mass range using the slider
- 1 indicates Success, 0 indicates Failure
- Color indicates Booster Version
  - Observed more launches for smaller payloads (compared to larger payloads)

Section 5

# Predictive Analysis (Classification)

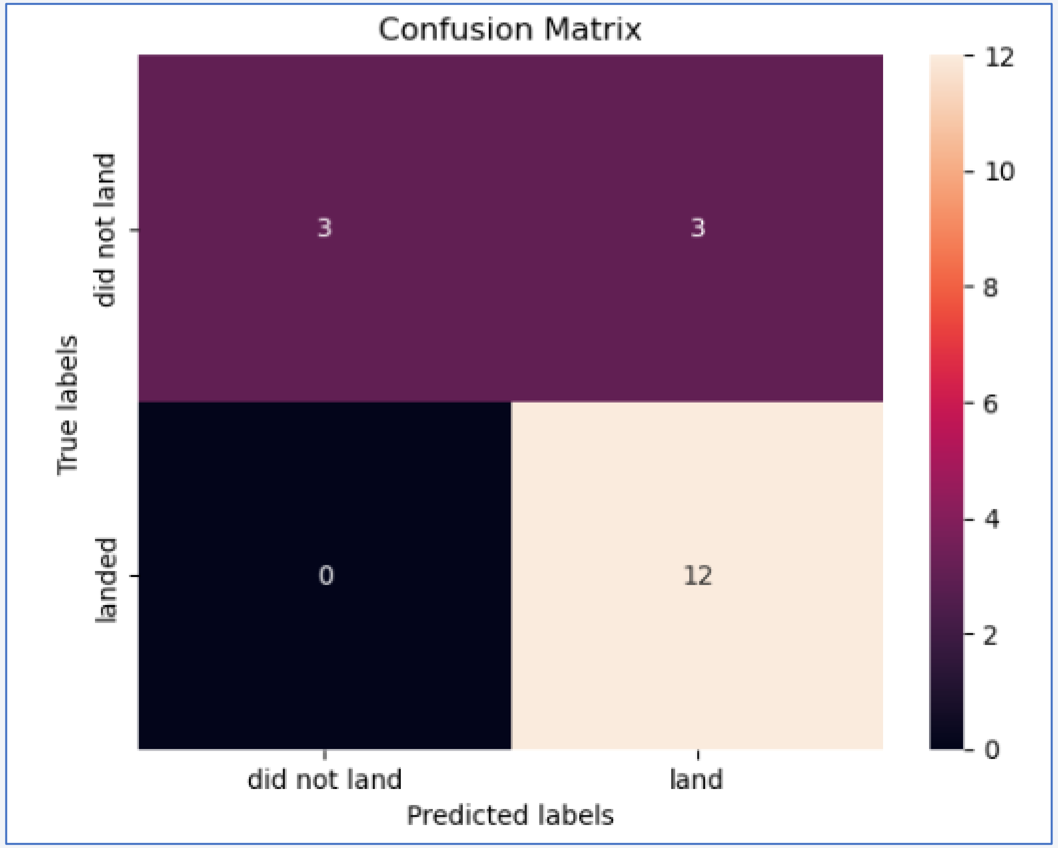
# Classification Accuracy

- **All models tested appear to perform with the same accuracy** (shown to right)
- Likely due to this being a small dataset



# Confusion Matrix

- All models showed the same **confusion matrix** due to having the same accuracy



# Conclusions

- Launch success was found to be related to a number of different factors
  - Launch success generally improved over time
  - Launches more frequently done with smaller payloads
  - Large payload launches are only done from specific Launch Sites
  - Success rates were better for certain orbit types
  - Payload seems dependent on the orbit type of the launch
  - Various models tested performed with similar accuracy, therefore any could be utilized for prediction
  - Launch Site location depends on variety of factors such as proximity to coastline, railways, and cities

# Appendix

- Associated scripts and files found here:
    - https://github.com/lbley24/data-science-course

# Thank you!